

Vivek Singh

Software Developer

vivek@myworkmails.com | (848) 235-3967 | USA | [LinkedIn](#) | [GitHub](#)

Summary

Skilled Software Developer with 4+ years in fullstack development, cloud technologies, and network architecture. Proficient in Spring Boot, Node.js, Express.js, AWS and database optimization. Experienced in building scalable, secure applications and automating processes to enhance performance. Strong problem solver and collaborative team player focused on delivering reliable, high-performance software solutions that align with business goals.

Technical Skills

- **Programming & Scripting:** Java, JavaScript (Node.js, React.js), Python, SQL (PostgreSQL, MySQL), NoSQL (MongoDB, Cassandra), Bash/Shell, TypeScript, YAML, JSON
- **Frameworks & Libraries:** Spring Boot (MVC, Security, Data JPA), Express.js, React.js (Redux, Hooks), Angular, Vue.js, Tailwind CSS, Bootstrap
- **Cloud, Infrastructure & IaC:** AWS (EC2, S3, RDS, Lambda, CloudFormation), Azure (App Service, Functions, DevOps), GCP (Compute Engine, Cloud Functions), Docker, Kubernetes (EKS, AKS), Terraform, Ansible
- **Databases, Messaging & Caching:** PostgreSQL, MySQL, MongoDB, Cassandra, Redis, Elasticsearch, Apache Kafka, RabbitMQ
- **DevOps, CI/CD & Monitoring:** Git/GitHub/GitLab, Jenkins, GitHub Actions, Azure Pipelines, Helm, Docker Compose, NGINX, Prometheus, Grafana, ELK Stack, Nagios
- **Security, Compliance & APIs:** OAuth2/JWT, AES-256 Encryption, PCI-DSS, IAM (AWS/Azure), Keycloak, HashiCorp Vault, RESTful APIs, GraphQL, SOAP, API Gateway, Webhooks

Work Experience

Software Developer, M&T Bank

10/2024 – Present | Remote, USA

- Developed and maintained a **Spring Boot**-based transaction processing and settlement system (**Java**, **Spring MVC**, **Spring Cloud**) that handles over millions daily transactions across ACH, wire transfers, and card payments, reducing end-to-end latency by 20 % and increasing throughput by 15 %.
- Exposed and documented **RESTful** endpoints using Spring MVC and Springdoc. Integrated with external payment networks via Apache HttpClient and **Spring Cloud LoadBalancer** to achieve a **99.99 % uptime SLA**.
- Enforced **AES-256** data encryption at rest (Spring Security Crypto) and implemented **JWT-based authentication/authorization** (Spring Security OAuth2), achieving full **PCI-DSS** compliance and eliminating all security audit findings.
- Tuned persistence layer with Spring Data JPA, Hibernate batch fetching, **PostgreSQL partitioning**, and Spring Cache (Caffeine), cutting average database response times by 10 % while managing TBs of transaction data.
- Architected and executed a comprehensive test suite with JUnit, Mockito, Spring TestContext, and Gatling for performance testing—simulating concurrent users to validate reliability, **fault tolerance**, and autoscaling policies.
- Built responsive front-end components with **React**, **Redux**, and **Tailwind CSS**. Integrated with **backend APIs** for real-time transaction dashboards, improving visibility and reducing manual error investigations by **15 %**.
- Managed source-code versioning with Git and GitHub: defined branching strategies, enforced pull-request reviews, and maintained release notes to streamline collaboration and ensure traceability.

Software Developer (Network), Capgemini

02/2021 – 12/2022 | India

- Designed a microservices-based network management platform using **Spring Boot** and Spring Cloud Netflix (Eureka, Ribbon) for service discovery and **client-side load balancing**, achieving **99.9 % network availability**.
- Implemented real-time network telemetry streaming with Spring for Apache Kafka and **Kafka Streams**. Created Prometheus exporters and **Grafana** dashboards to visualize key metrics, **reducing incident response times by 20 %**.
- Secured network management APIs and **VPN** services using Spring Security (OAuth2 Resource Server, method-level security) integrated with Keycloak and Vault. Enforced **RBAC** and encrypted credentials, cutting vulnerability scan issues by 25 %.
- Automated network provisioning workflows with Spring Integration (TCP adapters, transformers) and Ansible playbooks, achieving a 20 % reduction in manual configuration errors and accelerating rollout of new segments by 30 %.
- Developed a **React-based network monitoring** dashboard with **Redux** for state management, delivering real-time topology visualization and outage alerts to operations teams.
- Managed source-code versioning with Git. Established feature branches, conducted code reviews via pull requests, and maintained changelogs to ensure consistent quality and auditability.

Systems Engineer, ESDS

01/2018 – 01/2021 | India

- Developed enterprise web apps using **Node.js** and **RESTful APIs** with millisecond response times, **saving \$200K annually** through optimized cloud resource allocation, **containerization**, and ensuring high availability for scalable, cost-efficient infrastructure.
- Designed robust **Express.js routes** integrated with **MongoDB**, enabling efficient data exchange and dynamic user interactions, while optimizing front-end frameworks for faster load times, handling 1 million requests per minute with strong security and performance practices.
- Automated server health monitoring scripts and led migration from legacy systems to modern cloud infrastructure, reducing maintenance overhead by 50%, improving scalability, and proactively identifying and resolving performance bottlenecks for reliable, **data center operations**.

Project (Hackathon Winner)

- **Developed BraveBuddy, an AI-powered therapeutic companion for pediatric cancer patients; integrated Whisper for speech-to-text and GPT4o for personalized conversational responses, enabling real-time emotional support, reminders, and affirmations.**

Education

Master of Science in Computer Engineering

Stevens Institute of Technology

01/2023 – 12/2024 | NJ, USA

Bachelor of Engineering in Computer Engineering

Pune University

08/2017 – 05/2022 | Pune, India

Certificates

- **Advanced C++ (Udemy)**
- **AWS Solutions Architect (Udemy)**
- **Red Hat Fundamentals (Coursera)**
- **Shell Scripting (Coursera)**
- **Red Hat Certified Engineer (RHCE)**
- **Cisco Certified Network Associate (CCNA)**